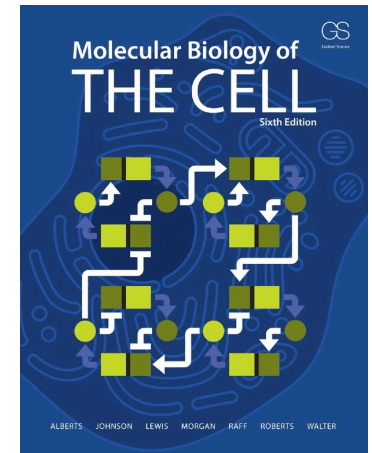


*Materials
Taken from*

Key concepts of Bioenergetics Part 1



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The Chemical Components of a Cell

atomic number

atomic weight

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	72 Hf
73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn				
87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	104 Rf
105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og				

Human body

Red (C, H, N, O) = 99%

Blue (Na, K, Mg, Ca, P, S, Cl)= 0.9%

Green = required in trace amounts.

Yellow = unclear whether essential.

* It seems: The chemistry of life is predominantly the chemistry of lighter elements.

Four Types of noncovalent attractions help Bring molecules Together in Cells

Three types of noncovalent bonds: electrostatic attractions (**ionic bonds**), **van der Waals attractions**, and **hydrogen bonds**

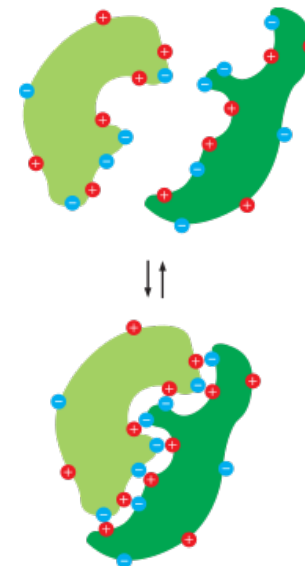
+

Fourth factor that can push molecules together: **the hydrophobic force**.

TABLE 2-1 Covalent and Noncovalent Chemical Bonds

Bond type		Length (nm)	Strength kJ/mole**	
			in vacuum	in water
Covalent		0.15	377 (90)	377 (90)
Noncovalent	ionic*	0.25	335 (80)	12.6 (3)
	hydrogen	0.30	16.7 (4)	4.2 (1)
	van der Waals attraction (per atom)	0.35	0.4 (0.1)	0.4 (0.1)

*An ionic bond is an electrostatic attraction between two fully charged atoms. **Values in parentheses are kcal/mole. 1 kJ = 0.239 kcal and 1 kcal = 4.18 kJ.



Electrostatic attractions are shown, in reality **all four noncovalent forces often contribute** to holding two macromolecules together

Noncovalent attraction (H-bond, van der Waals) would be much too weak to be effective in the face of thermal motions, their **energies can sum to create a strong force** between two separate molecules.

Cells Contain four major families of small organic molecules

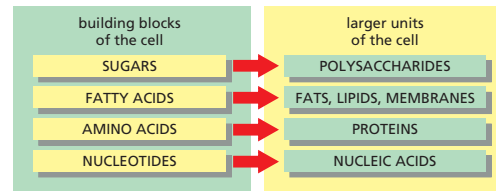
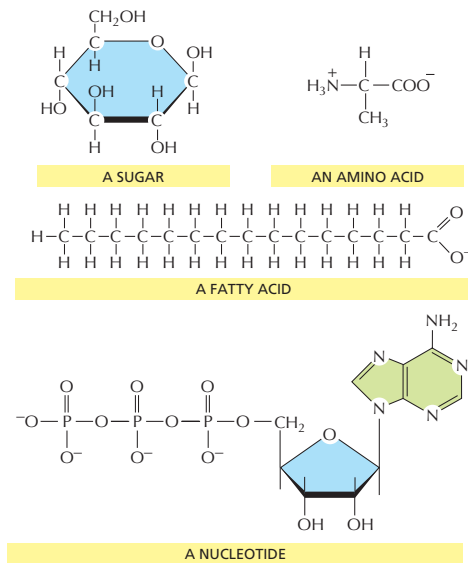


Figure 2-6 The four main families of small organic molecules in cells. These small molecules form the monomeric building blocks, or subunits, for most of the macromolecules and other assemblies of the cell. Some, such as the sugars and the fatty acids, are also energy sources. Their structures are outlined here and shown in more detail in the Panels at the end of this chapter and in Chapter 3.

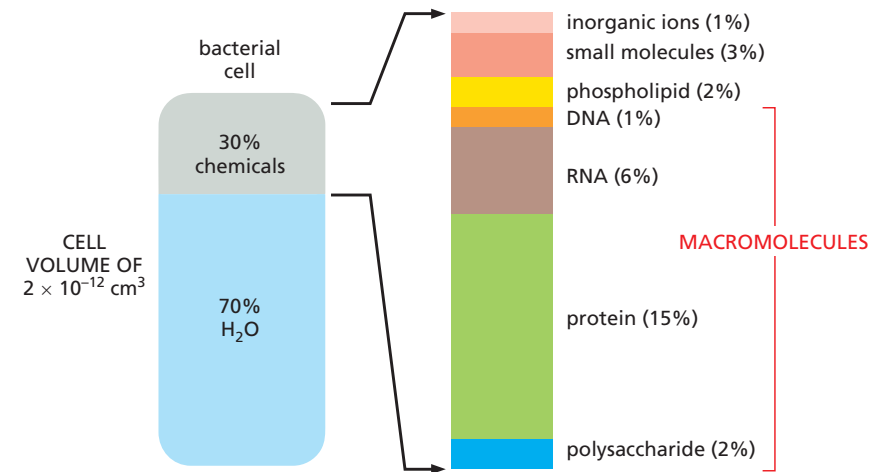


Figure 2-7 The distribution of molecules in cells. The approximate composition of a bacterial cell is shown by weight. The composition of an animal cell is similar, even though its volume is roughly 1000 times greater. Note that macromolecules dominate. The major inorganic ions include Na^+ , K^+ , Mg^{2+} , Ca^{2+} , and Cl^- .

Three families of macromolecule formed by polymerization

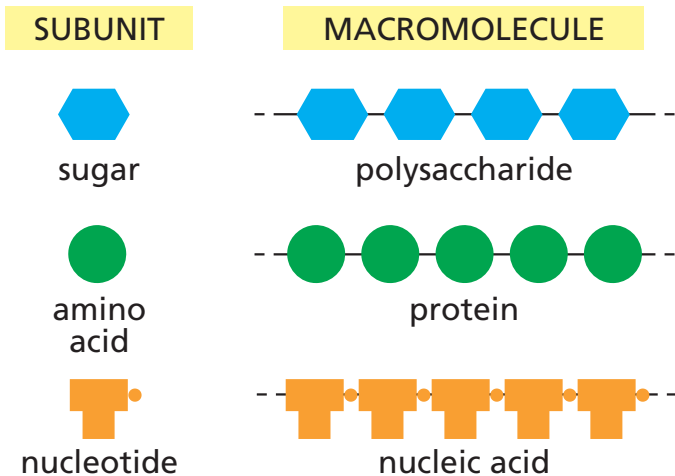


Figure 2–8 Three families of macromolecules. Each is a polymer formed from small molecules (called monomers) linked together by covalent bonds.

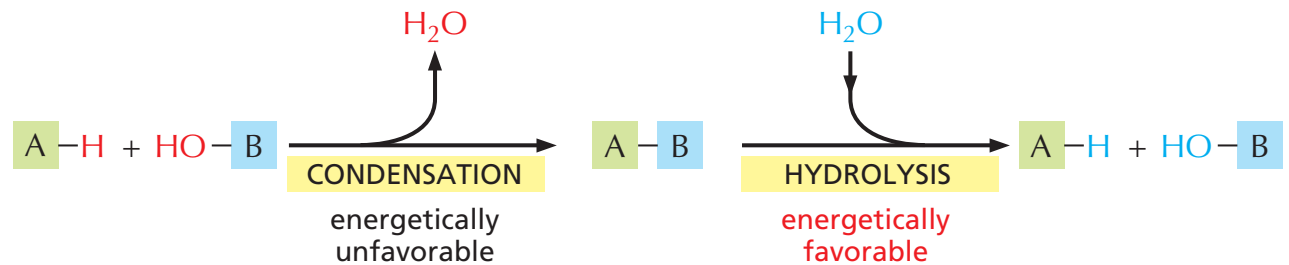


Figure 2–9 Condensation and hydrolysis as opposite reactions. The macromolecules of the cell are polymers that are formed from subunits (or monomers) by a condensation reaction, and they are broken down by hydrolysis. The condensation reactions are all energetically unfavorable; thus polymer formation requires an energy input, as will be described in the text.

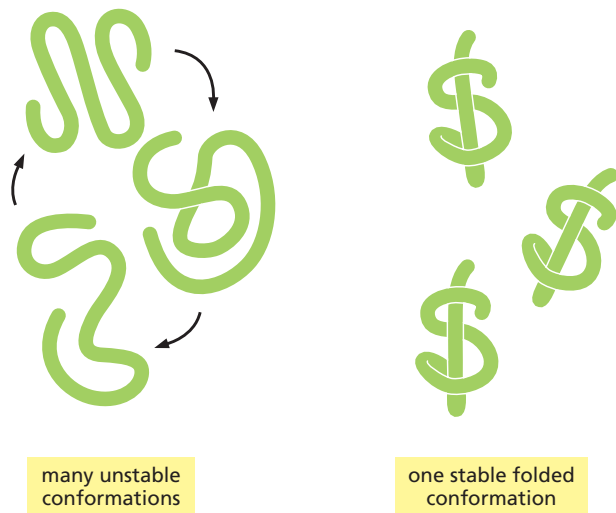


Figure 2–10 The folding of proteins and RNA molecules into a particularly stable three-dimensional shape, or conformation. If the noncovalent bonds maintaining the stable conformation are disrupted, the molecule becomes a flexible chain that loses its biological activity.

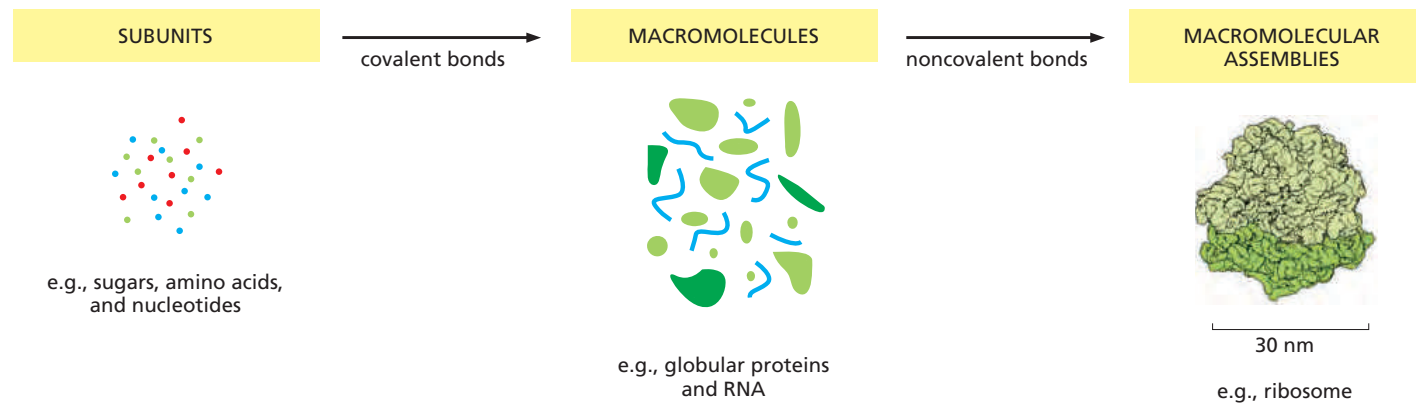


Figure 2–11 Small molecules become covalently linked to form macromolecules, which in turn assemble through noncovalent interactions to form large complexes. Small molecules, proteins, and a ribosome are drawn approximately to scale. Ribosomes are a central part of the machinery that the cell uses to make proteins: each ribosome is formed as a complex of about 90 macromolecules (protein and RNA molecules).

Summary

- * Living organisms = autonomous, self-propagating chemical systems.

- * Composed of restricted set of small carbon-based molecules

- * Molecules are made-up of covalent bonds (main categories: sugars, fatty acids, amino acids, and nucleotides).

Sugars : primary source of chemical energy for cells and stored as polysaccharides for energy storage.

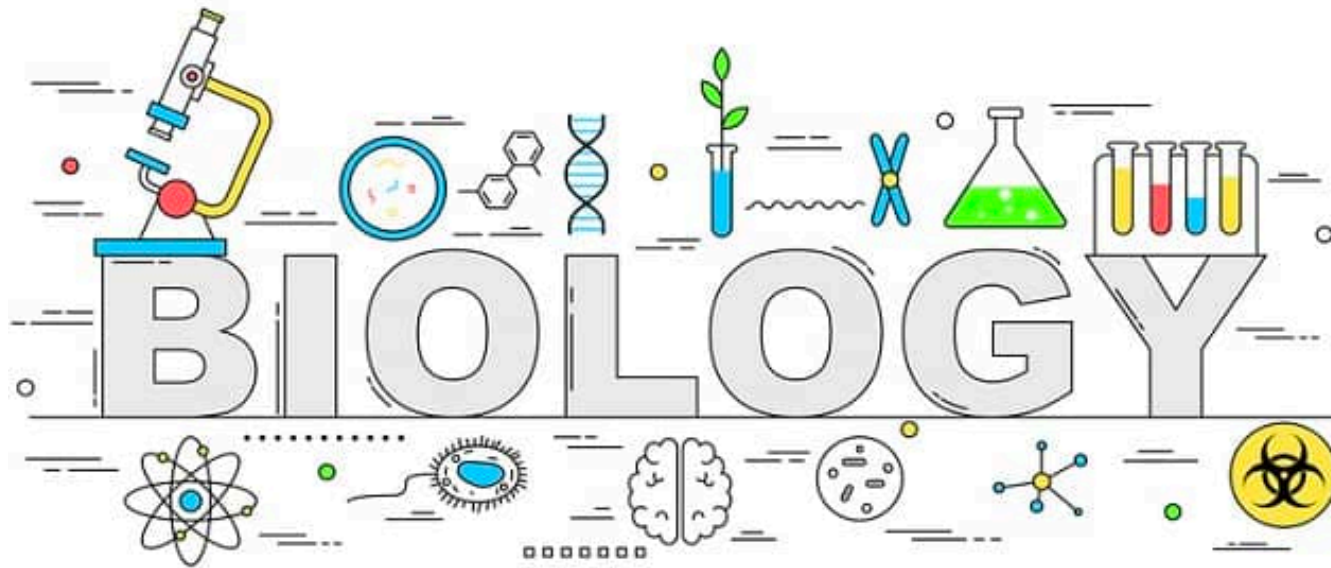
Fatty acids: Structural role (formation of cell membranes) + storage of energy.

Amino acids: Proteins (Structural, biological catalyst)

Nucleotides: Informational macromolecules, RNA and DNA.

- * Macromolecules (Protein and RNAs) fold into a unique conformation (key for its function). This folding creates unique surfaces, and it depends on a large set of weak attractions produced by noncovalent forces between atoms.

- * Macromolecules assemble by non-covalent forces and form large complexes which may have specific function in cells.



Next: Key concepts of Bioenergetics (Part 2)